

From the real ant to the artificial ant

Applications in combinatorial optimisation, data clustering, collective robotics and image processing

Natural algorithms

Nature-inspired algorithms help forensic investigators crack complex cases

From the labs

virtual network. The TCP transport layer running over IP ensures that large streams of data traffic are transported reliably, in the right order. All applications such as FTP and the WWW are layered on top of TCP, and thus we obtain our multilayered interacting network. In this network, the dynamics of decentralized packet transfer can respond to congestion protocols in the higher levels invoked by requirements generated at the application layers. The result is a complex tiered network responding to feedback loops between the various tiers.

Butterfly wings and peacock feathers create color by causing light to interfere with itself. These highly developed structures reflect light so that specific wavelengths interfere with each other to create nature's purest, most vivid colors. This inspired Qualcomm's next generation of display technology, Interferometric Modulator (IMOD) that achieves substantial performance improvement and power-saving benefits over current technologies. This innovative, highly reflective design enables an Always-On[®] experience with:

- ❑ Ultra-low power consumption — extended battery life
- ❑ Superior viewability — in all lighting conditions, even bright sunlight
- ❑ Enhanced readability — reduced eye fatigue, like reading printed paper
- ❑ Robust functionality — virtually unaffected by environment and extreme temperatures
- ❑ Mechanical durability — long operational life

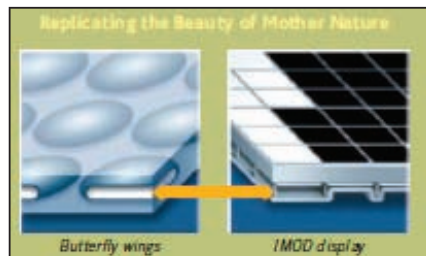
Biomolecular computing

There have been various Molecular electronics attempts to mimic solid-state components with molecular structures. Molecular wires, rectifiers, and transistors have all been constructed on very small scales. Biomolecules such as the protein bacteriorhodopsin have shown great promise for natural computing. This protein



Some of nature's creations depict vivid colours

serves as a solar power source for a bacterium. Light at a certain wavelength can change how the protein behaves; this behaviour is accompanied by a colour change. This lets us, in principle, encode states and then read them off. This biomolecule and others like it might have



Qualcomm's IMOD displays are inspired from butterfly wings

serious applications in high-density memory. Researchers have already used a genetically engineered version to build devices that can perform optical character recognition.

IBM increased chip performance, courtesy shells

Thanks to the way shells balance heat when under high pressure, IBM has come out with its own Airgap Technology.

Though putting a bigger fan and copper rods is a good solution, it increases the amount of noise, power consumption and energy loss. Lenovo overcame one of these problems with an idea inspired by owls to lower the sound. Owls fly swiftly without making any noise to catch its prey. Similarly, Lenovo fans run faster and without significant increase in noise by changing the shape of the fan blades.

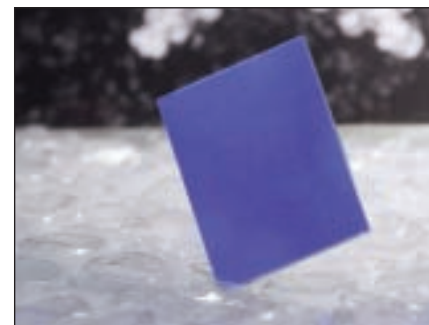
Bionic device technologies

Biological models of the vertebrate, insect and mammalian visual systems have led to several bio-inspired algorithms. Many man-made electro-optical materials and devices have been inspired by biological systems. The motion detection system of the fly is a good example of a useful optoelectronic system for robot piloting and navigation. The rest of the insect world represents a huge data base for future bio-inspired microsensor-processor-actuator units.

Naturally, the future growth of VLSI faces a number of limitations. But, you will be pleased to know that Biology has been employed to overcome some of them, an example being fault tolerance, by using loosely coupled, faulty

elements. Although mimicking biology with silicon is very hard. The real niche for bio-inspired systems is at the interface between the digital and analogue world, particularly in the areas of wireless telecommunication and PDAs where power dissipation is critical for implementing powerful signal processing applications like speech recognition. Current digital trends are toward full integration on a single chip. However, many systems with high dimensional input data need the massive processing of analogue signal arrays.

New nano-scale electronic interfaces will offer great improvements in ultra-low-power bionic devices for prosthesis and scientific studies of neural tissue. Future advances in microelectronics and analogue processing will allow dramatic reductions in cost and size. Advanced imaging sensors can benefit greatly from new bio-inspired algorithms.



IBM's chip technology is inspired from shells

The future

We envision that within a decade, and in a continuously developing fashion, a new industry will emerge thanks to research inspired by nature that will serve unprecedented human needs. It's not far before we have sensory and other prostheses (visual, auditory, tactile, olfactory, gustatory, motion based, etc.), as well as self-contained medication, forecasting and therapeutic devices built into the human body or brain-controlled robots and motor prostheses.

With some effort by scientists worldwide, we will be living in a world with life-like perception/action devices for household, industrial, and medical use, and OEM bionic parts to be embedded in many other goods and services. It will be a networked world with brain-controlled, neuromorphic or bio-inspired robots. Scary as it may sound, with nature on our side, the future beckons with open arms! 🦾